



Short Communication

Deep Learning Forecasts the Occurrence of Sleep Apnea from Single-Lead ECG

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(Received 17 November 2021; accepted 15 February 2022; published online 17 March 2022)

Abstract

Objectives—Sleep apnea is the most common sleep disorder that leads to serious health complications if not treated early. Forecasting apnea occurrence ahead in time provides the opportunity to take appropriate actions to control and manage it.

Methods—A novel framework for forecasting the occurrence of apnea from single-lead electrocardiogram (ECG) based on deep recurrent neural networks is proposed. ECG R-peak amplitudes and R-R intervals are extracted and aligned using power spectral analysis, and recurrent deep learning models are developed to extract the most predictive ECG features and forecast the occurrence of apnea.

Results—The performance of the proposed approach was validated in forecasting apnea events up to five minutes in future on a dataset of 70 sleep recordings. A forecasting accuracy of up to 94.95% was achieved which was higher than the performance of conventional multilayer perceptron ($p < 0.05$) and other state-of-the-art techniques.

Conclusions—The proposed deep learning approach was successful in forecasting the occurrence of sleep apnea from single-lead ECG. It can therefore be adopted in wearable sleep monitors for the management of sleep apnea. Our developed algorithms are publicly available on GitHub.

Keywords—Deep learning, Electrocardiogram, Forecasting, Recurrent neural network, Sleep apnea.

INTRODUCTION

Sleep apnea, the cession of breathing during sleep, impacts nearly 1 billion people worldwide and its prevalence is increasing.⁵ Sleep apnea has a detrimental impact on sleep and quality of life and can lead to diabetes, depression, hypertension, and memory loss if

not treated.^{7,21} In this context, forecasting the occurrence of sleep apnea ahead in time provides the opportunity to take appropriate actions to manage, control, and treat it.

Detection of sleep apnea from different physiological signals has been widely studied.^{4,13,15} Several mathematical, statistical, and machine learning models have been developed to quantify sleep and to detect the occurrence of sleep events such as apnea by analyzing the physiological signal patterns during the corresponding event.^{2,3,11,22} However, forecasting the occurrence of sleep apnea has not yet been adequately investigated. In Ref. 12, a nonparametric mixture-gaussian-process model was developed to estimate the transition from normal states to an obstructive sleep apnea state using ECG-derived HRV parameters. In Ref. 17, time-series features were extracted from the nasal airflow signal and several conventional machine learning algorithms including support vector machine (SVM), logistic regression, and k-nearest neighbor were used for sleep apnea forecasting. It was found that SVM achieves the best forecasting performance. In Ref. 26, a neural network was developed to forecast the occurrence of sleep apnea events up to 30 seconds into the future from EEG, ECG, EOG, EMG, nasal pressure, and oronasal temperature. In Ref. 14, several architectures of neural networks were investigated for forecasting the occurrence of sleep apnea from nasal airflow and abdominal and thoracic movement signals. The best performance was obtained by a multilayer perceptron (MLP). In Ref. 6, an evolutionary algorithm was utilized to derive a set of IF-THEN rules from ECG-derived HRV parameters in order to forecast the occurrence of sleep apnea. In Ref. 16, continuous wavelet and short-time Fourier transforms were used to represent 30-s ECG segments by their

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scalogram and spectrogram, respectively. AlexNet, GoogleNet, and ResNet18 deep models were used to predict the apnea events in future. Best prediction results were achieved using scalograms and a modified ResNet18 immediately 30 s before apnea onset with accuracy of 82.3%.

Many of the existing studies on forecasting the occurrence of sleep apnea rely on the analysis of multiple physiological signals that require cumbersome instruments^{14,26} or physiological signals that are difficult to measure such as nasal air flow.¹⁷ In addition, many studies use conventional machine learning algorithms that rely on expert-defined features.^{6,11,12,14,17,26} Such hand-crafted features may not fully capture the complex information hidden in the time pattern of physiological signals. Given the complexity of human physiological dynamics and the variabilities due to different internal events and external stimuli, advanced predictive algorithms are needed that can capture physiological patterns indicative of the occurrence of sleep apnea into the future.

Here, we demonstrate how deep learning can be employed to reliably forecast the occurrence of sleep apnea episodes ahead in time using single-lead ECG data from the past and present, i.e., before clinical symptoms such as stop breathing occur. We propose a new deep recurrent framework to capture the time pattern of ECG signal several minutes before the occurrence of sleep apnea. Power spectral analysis is performed to align ECG segments and deep recurrent networks are used to automatically extract spectral features from short-time intervals and to model the spectral features evolution over longer-time periods. The algorithms are solely based on the analysis of single-lead ECG-derived R-R intervals and R-peak amplitudes that contain important information about the cardiorespiratory function and can be easily recorded by wearables. Unlike conventional machine learning algorithms that rely on expert-selected features, the proposed approach automatically extracts the most predictive sleep apnea features. The algorithms are validated on 70 sleep recordings from 32 individuals and compared with the conventional neural networks as well as the state-of-the-art techniques in forecasting the occurrence of sleep apnea. We have made our algorithms publicly available on GitHub at <https://github.com/mahsaabahrami/Forecasting-Sleep-Apnea>.

METHODS

Experimental Design

Physionet Apnea-ECG Database v1.0.0²⁰ was used in this study. It consists of 70 ECG sleep recordings collected from 32 individuals (7 females, age: 44 ± 11 , ap-

nea-hypopnea index, AHI: 24 ± 25). The length of recordings was 8.2 ± 0.52 h. In the database, sleep apnea annotations, labeled by experts, are available for every 1 min. In the expert evaluation, each 1-min segment is evaluated in a binary way as normal or apnea. The total number of 1-min segments are 34,428 consisting of 21,362 segments of sleep apnea and 13,066 segments with no apnea event. The designed forecasting algorithm were accordingly designed to forecast the class label of each 1-min segment in future in a binary. Further details on the measurement protocol and participants' characteristics can be found in Ref. 20.

Preprocessing and Data Preparation

ECG signal, digitized at 100 Hz, was first denoised using a bandpass filter with lower and upper cutoff frequencies of 0.1 and 35 Hz. Automatic algorithms were used to detect ECG R peaks and compute R-R intervals and R-peak amplitudes. ECG R-R intervals and R-peak amplitudes were evenly interpolated at 3 Hz.

Given that apnea labels are provided for every 1 min, the ECG R-R intervals and R-peak amplitudes were divided into non-overlapping 1-min segments (here, we call them fast-time data) of length 180 samples (3 Hz sampling rate). To model the time evolution of ECG pattern over longer time periods (here, we call them slow-time data), n successive 1-min segments were analyzed. This resulted in a 3D data structure of $180 \times 2 \times n$ dimension, as shown in Fig. 1.

Power Spectral Analysis

Given that our physiological time series data originate from cardiac activity that may not happen at the exact same time and with the same rate over the analysis period, they need to be aligned. Data temporal alignment facilitates the learning process of the model without requiring a huge amount of training data.²⁴

Here, to address the temporal alignment issue, we transformed the data into frequency domain. Unlike time domain, the signal samples in frequency domain are aligned where all the data samples at the same frequency bin represent the signal energy at that specific frequency. We calculated the power spectral density (PSD) using Welch's method. This led to a 3D data structure of $180 \times 2 \times n$ dimension, where the fast-time data were represented in frequency domain.

Deep Learning

To extract the most predictive spectral features from the frequency-transformed fast-time data and model the time evolution of these features over longer time

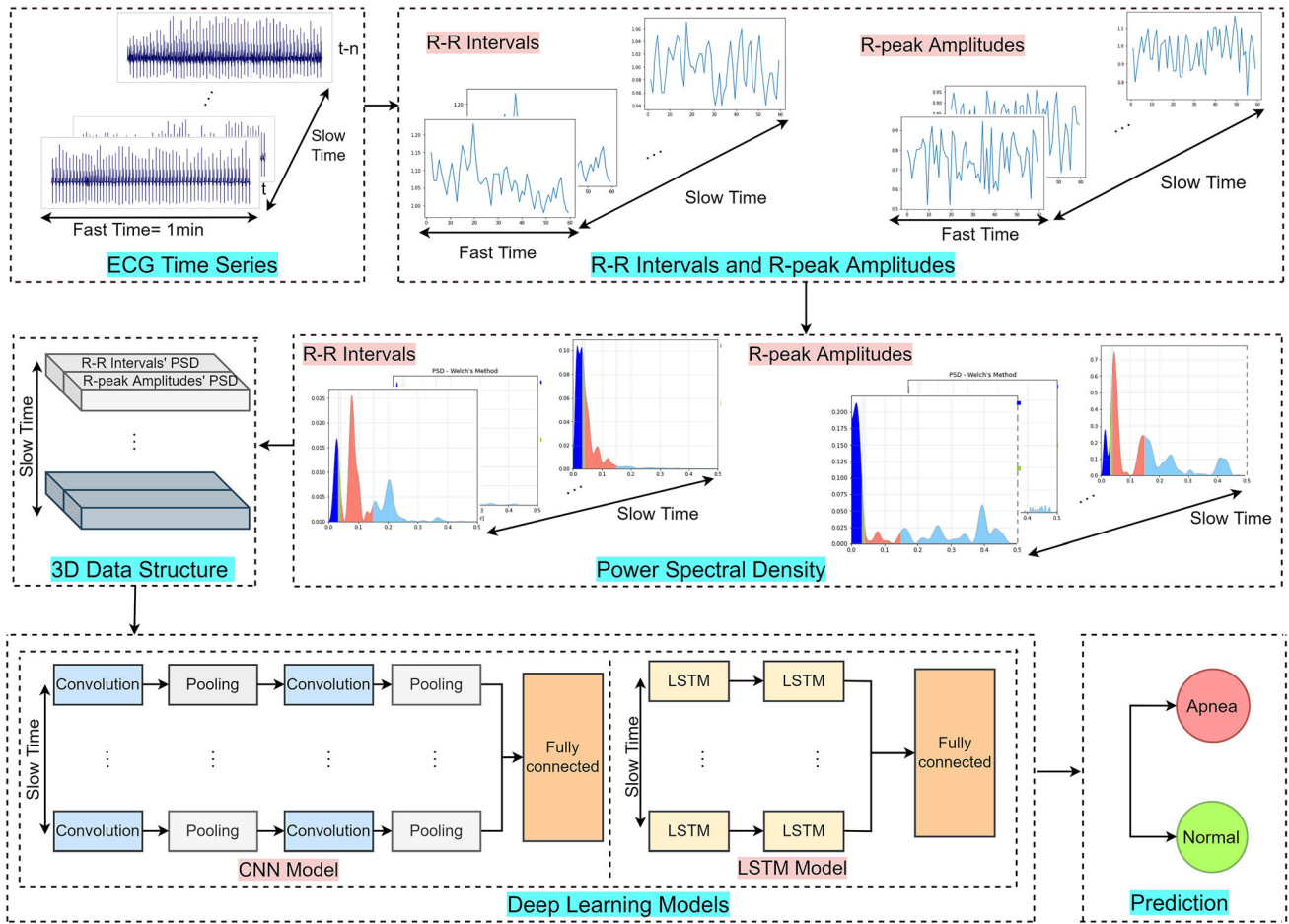


FIGURE 1. Block diagram representation of the proposed deep recurrent framework for forecasting the occurrence of sleep apnea from single-lead ECG.

periods (slow time) prior the occurrence of sleep apnea, two architectures of deep neural networks were developed.

Convolutional Neural Network (CNN)

A convolutional neural network was designed to extract features from the frequency-domain fast-time data and a fully connected layer was used to forecast the occurrence of sleep apnea by capturing the time evolution of the extracted features over the slow time. The developed CNN comprised of 32 kernels of size 5 slid over the frequency-domain fast-time data followed by a 3×1 max-pooling operation in the first layer (leading to output dimension of $58 \times 32 \times n$), and 64 kernels of size 5 followed by a 3×1 max-pooling in the second layer²⁵ (leading to output dimension of $18 \times 61 \times n$). Data were then flattened and fed to two fully connected layers with $\sqrt{(18 \times 64 \times n)}$ and $\sqrt{(18 \times 64 \times n)}$ nodes, respectively, followed by a SoftMax layer to forecast apnea events.

Long Short-Term Memory (LSTM)

LSTM network has been particularly designed for modeling time series.¹⁰ Here, we designed a 2-layer n -cell stack LSTM network where each cell received a 2D structure of 180×2 . Four nodes were used in each LSTM cell leading to a $4 \times n$ output dimension. A fully connected layer with $\sqrt{(4 \times n)}$ nodes followed by a SoftMax layer was then used to forecast apnea events.

RESULTS

The Physionet Apnea-ECG Database consisting of 70 sleep records was used in this study. For a reliable and accurate analysis, we performed a 10-fold stratified cross-validation. By performing cross-validation, we're able to use all the data for training and for testing while evaluating our learning algorithm on examples it has never seen before. Using stratified cross-validation, we ensured that each fold contains roughly the same proportions of the two types of ap-

nea and non-apnea classes. In each fold, 10% of training data was used as a validation set to select model hyperparameters and avoid overfitting. The results were reported on the test data not seen during training and validation. The performance of the proposed recurrent framework using deep learning algorithms (CNN, LSTM) as well as the conventional MLP

was evaluated in forecasting the occurrence of sleep apnea episodes in the 1st to 5th min into the future.

Table 1 summarizes the performances of different algorithms in terms of accuracy, sensitivity, specificity, and F-score, using slow-time lengths of $n = 5, 10, 20$ min, on the test data not seen during training and validation. It is observed that LSTM network achieved superior results in terms of accuracy (94.95%), sensi-

TABLE 1. Performance comparison between different learning models in the proposed apnea forecasting framework.

Model	Time in Future [min]	Accuracy (%)	Sensitivity (%)	Specificity (%)	F-Score (%)
Slow time $n = 5$ min					
MLP	1	83.44 $\pm$ 0.37	79.27	86.08	78.77
	2	81.92 $\pm$ 0.50	77.33	84.84	76.85
	3	80.85 $\pm$ 0.70	74.84	84.66	75.23
	4	79.44 $\pm$ 0.81	73.89	82.98	73.67
	5	78.22 $\pm$ 0.70	73.67	81.13	72.51
CNN	1	88.12 $\pm$ 0.63	82.87	91.44	84.38
	2	86.66 $\pm$ 0.68	82.20	89.48	82.70
	3	85.66 $\pm$ 0.35	81.73	88.17	81.60
	4	84.69 $\pm$ 0.49	79.25	88.16	80.12
	5	84.02 $\pm$ 0.66	78.71	87.43	79.36
LSTM	1	91.38 $\pm$ 0.71	88.78	93.03	88.78
	2	90.54 $\pm$ 0.39	88.52	91.83	87.91
	3	89.63 $\pm$ 0.35	87.30	91.10	86.75
	4	88.72 $\pm$ 0.58	86.83	89.92	85.71
	5	88.10 $\pm$ 0.91	85.63	89.68	84.88
Slow time $n = 10$ min					
MLP	1	85.61 $\pm$ 0.48	81.34	88.35	81.56
	2	83.04 $\pm$ 1.02	77.91	86.35	78.24
	3	82.22 $\pm$ 1.26	77.28	85.40	77.33
	4	80.88 $\pm$ 0.89	75.40	84.43	75.61
	5	79.93 $\pm$ 0.73	73.04	84.41	74.11
CNN	1	88.61 $\pm$ 0.45	84.85	91.02	85.34
	2	87.16 $\pm$ 0.23	82.63	90.08	83.44
	3	86.24 $\pm$ 0.52	81.96	89.00	82.38
	4	85.19 $\pm$ 0.60	81.75	87.43	81.27
	5	84.64 $\pm$ 0.50	80.65	87.23	80.51
LSTM	1	94.18 $\pm$ 0.61	93.24	94.80	92.61
	2	93.51 $\pm$ 0.36	92.47	94.18	91.78
	3	93.12 $\pm$ 0.32	92.09	93.78	91.30
	4	92.66 $\pm$ 0.41	91.58	93.36	90.74
	5	92.54 $\pm$ 0.53	91.52	93.20	90.62
Slow time $n = 20$ min					
MLP	1	86.41 $\pm$ 0.78	82.46	89.01	82.80
	2	84.74 $\pm$ 0.67	80.03	87.84	80.61
	3	83.51 $\pm$ 0.74	78.82	86.60	79.14
	4	82.39 $\pm$ 0.97	77.55	85.59	77.77
	5	81.55 $\pm$ 0.78	77.45	84.27	76.97
CNN	1	88.36 $\pm$ 0.42	84.41	90.96	85.18
	2	87.08 $\pm$ 0.37	83.36	89.53	83.36
	3	86.24 $\pm$ 0.65	81.09	89.63	82.39
	4	85.93 $\pm$ 0.25	80.97	89.20	82.06
	5	84.95 $\pm$ 0.66	79.40	88.58	80.73
LSTM	1	94.95 $\pm$ 0.33	93.92	95.63	93.65
	2	94.32 $\pm$ 0.51	93.28	95.00	92.88
	3	94.02 $\pm$ 0.48	93.08	94.63	92.52
	4	93.56 $\pm$ 0.56	92.67	94.14	91.97
	5	93.23 $\pm$ 0.37	92.28	93.86	91.56

tivity (93.92%), specificity (95.63%), and F-score (93.65%) compared to the rest of the networks. CNN network was superior to the conventional MLP but was outperformed by LSTM. A set of corrected paired-sample t tests confirmed that the CNN and LSTM improved accuracies were statistically significant compared to the conventional MLP ($p < 0.05$).

The achieved results demonstrate the power of deep learning approach in extracting the most representative ECG spectral features and as well as capturing the ECG temporal patterns pinpointing the occurrence of sleep apnea in future.

DISCUSSION AND CONCLUSIONS

By the development of smart sleep wearables capable of unobtrusive monitoring different physiological signals, the accurate forecasting of sleep events such as apnea is becoming more feasible. Forecasting the occurrence sleep apnea can help in devising appropriate preventive interventions as well as therapies and management strategies. Here, we proposed a general framework for modeling ECG physiological pattern prior the occurrence of sleep apnea using spectral analysis and deep learning. Spectral analysis facilitated data alignment and deep learning enabled the automatic extraction of features as well as physiological pattern modeling and recognition. It was found that LSTM-based models are superior to other neural network architectures in forecasting of sleep apnea occurrence. This can be attributed to the better capability of LSTM in modeling time series data.

It was observed that the performances of the proposed methods slightly degrade when forecasting events further in future. For example, LSTM's per-

formance was degraded by about 2% in detecting events occurred 5 mins into the future compared to 1 min. In most cases, the best results were achieved when forecasting apnea events 1 min into the future.

Different slow-time data lengths ($n = 5, 10, 20$ min) were considered in this study and best results were achieved for a slow-time length of $n = 20$ min in the past as reported in Table 1. For example, forecasting accuracy of LSTM network when predicting the occurrence of apnea in the next 1 min, was improved 2.8% when analyzing the last 10 min instead of 5 min ($p < 0.05$), and 0.77% when analyzing the last 20 min instead of 10 min ($p < 0.05$). Considering more data in the past may further improve the prediction performance; however, it will lead to a more complex network with more parameters to be tuned.

Unlike methods that rely on the analysis of physiological signals that are not easy to measure,^{14,17,26} our method relies on a single-lead ECG and therefore can be easily employed in wearables. Unlike the conventional machine learning methods that rely on hand-crafted features,^{6,11,12,14,17,26} our deep model processes the raw R-R interval and R-peak amplitude signals and is therefore capable of automatic extraction of the most informative features. Unlike deep learning methods that rely on the analysis of 2-D transformed data using complex models such as ResNet18,¹⁶ our proposed method is of lower complexity. It only requires the analysis of R-R interval and R-peak amplitude signals using a 2-layer stack LSTM network with significantly fewer parameters. Table 2 compares our proposed approach with state-of-the-art techniques in terms of performance and complexity. It can be observed that our proposed approach outperforms the rest of the algorithms. Given that the reported results in Table 2 are obtained on different datasets,

TABLE 2. Comparison of different apnea forecasting algorithms.

Method	Dataset	Time in future (s)	Best accuracy	Best sensitivity (%)	Best specificity (%)	Number of trainable parameters
Time-series features + SVM ¹⁷	Nasal airflow from 13 individuals	30	86.90%	90.60	76.30	31
Wavelet features + LAMSTAR ²⁶	EEG, ECG, EOG, EMG, nasal pressure, and oronasal temperature from 73 individuals	30 ^a , 60, 90, 120	–	80.60	72.80	4410
Spectrogram + Modified ResNet18 ¹⁶	Single-lead ECG from 152 individuals (HomePap and ABC datasets ²⁷)	30 ^a , 60, 90	80.13%	81.99	77.25	9,354,114
Scalogram + Modified ResNet18 ¹⁶	Single-lead ECG from 152 individuals (HomePap and ABC datasets ²⁷)	30 ^a , 60, 90	82.30%	83.22	82.27	9,354,114
PSD + LSTM (Proposed method)	Single-lead ECG from 32 individuals (Physionet Apnea-ECG Dataset ²⁰)	60 ^a , 120, 180, 240, 300	94.95%	93.92	95.63	30,013

^aBest results achieved.

further study is required to comprehensively compare different machine learning and deep learning methods for sleep apnea forecasting.

Our proposed framework was based on the analysis of ECG R-R intervals and R-peak amplitudes using deep learning. In order to verify the contribution of R-R intervals and R-peak amplitude, the proposed approach was once implemented using only the R-R intervals as input and once using the R-peak amplitudes as the input to the deep models. We found that the best forecasting accuracy (when analyzing $n = 20$ min of past data to predict apnea events 60 s in future using LSTM model) is dropped from 94.95 to 92.59% by feeding solely the R-R intervals and to 88.57% by feeding solely the R-peak amplitudes to the LSTM model. This shows that the R-R intervals are more informative and have more predictive value compared to the R-peak amplitudes for sleep apnea forecasting. The combination of R-R intervals and R-peak amplitudes achieves the best accuracy.

The capability of the proposed algorithm to forecast a sleep apnea episode with a temporal horizon of 1 to 5 min depends both on input data size and the system computational power. When analyzing 20-min slow-time data in the past, the forecasting time was 1.03 and 0.41 s for CNN and LSTM networks, respectively. All the algorithms were developed in Python 3.8 on a Windows 10 PC with Intel Core i7-5500U CPU @2.40GHz and with 8GB RAM. The proposed method has therefore the potential for real-time processing and forecasting.

In this work, the focus was on forecasting the occurrence of apnea events by analyzing the R-peak amplitude and R-R intervals extracted from a single-lead ECG. Respiratory and circulatory systems are interconnected thorough the autonomic nervous system regulation. During inhalation, the intra-thoracic pressure decreases which in return lowers the atrial pressure and decreases the activation of baroreceptors. This results in reduced vagal tone and increased heart rate (decreased R-R intervals). During exhalation, the intra-thoracic pressure increases due to relaxation and upward movement of the diaphragm. As a result, atrial pressure increases, and the baroreceptors are activated. This leads to increased vagal tone and decreased heart rate (increased R-R intervals).⁹ The change in orientation of the electrical axis of the heart during respiration, relative to ECG electrodes, induces an amplitude modulation effect on the ECG signal (change in R-peak amplitude).¹⁸ When an apnea event occurs, the blood oxygen level starts to decrease, and the cardiorespiratory system tries to compensate this thorough the change in heart rate as a reflection of sympathetic and parasympathetic activity.¹⁶ Therefore, it is expected that sleep breathing disorders can be

detected by studying cyclical variation of heart rate (ECG R-R intervals) combined with respiration-modulated changes in ECG morphology (ECG R-peak amplitude).¹⁹ This is in accordance with previous studies where features based on the timing of QRS complexes and the amplitude of the ECG have been found to be useful for apnea identification.¹

The dataset used in our study included 19 individuals diagnosed with sleep apnea ($AHI > 5$) and 13 normal individuals ($AHI < 5$). The proposed approach forecasts the occurrence of individual sleep apnea episodes in both groups based on the past and present pattern of the ECG signal. Therefore, the proposed approach can be used for forecasting the occurrence of sleep apnea events in patients diagnosed with apnea as well those that are yet undiagnosed.

Given the complexity of physiological temporal patterns due to different sources of variability, future work can be directed toward the adoption attention mechanism²³ in the proposed deep learning framework to enhance the important parts of physiological data and fading out the rest. This will also improve model interpretability by highlighting the most important temporal features.

A limitation of this study was the lack of large sample size. Evaluation of the proposed method on a larger sleep dataset can further demonstrate the performance of the proposed deep learning framework.

The developed forecasting framework only relied on a single-lead ECG. A single channel ECG is sensitive enough to detect ECG R-peak amplitudes and R-R intervals.⁸ This can be particularly useful in the design of inexpensive and easy-to-use sleep wearables that can reliably monitor the occurrence of sleep apnea. As a future work, the proposed approach can be further extended for the detection and forecasting of other sleep events and thereby facilitate sleep and health management.

ACKNOWLEDGMENTS

The work of Mohamad Forouzanfar was supported in part by the Natural Sciences and Engineering Research Council of Canada (NSERC, RGPIN-2021-03924). The content is solely the responsibility of the authors and does not necessarily represent the official views of the Natural Sciences and Engineering Research Council of Canada.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

RESEARCH INVOLVING HUMAN AND ANIMAL RIGHTS

No human studies were carried out by the authors for this article. No animal studies were carried out by the authors for this article.

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